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NeuroClean: High-Fidelity Smartphone Image Denoising via Hybrid U-Net–DnCNN Dual-Stream Deep Learning and Overlapping Hann-Window Tiling

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ABSTRACT Smartphone cameras operate under acute physical form-factor limitations: micro-apertures and sub-micron to micron pixel pitches (1.0 μm – 1.4 μm) restrict photon capture, yielding severe Poisson photon shot noise, amplifier read noise, and chromatic grain under low-ambient illumination. Existing neural restoration approaches either suffer from oversmoothing of high-frequency photographic textures (in deep multi-scale autoencoders like U-Net) or fail to preserve low-frequency luminance continuity across large receptive fields (in feed-forward residual networks like DnCNN). Furthermore, deploying deep models on arbitrary, high-resolution smartphone captures (12+ to 100+ megapixels) induces fatal memory exhaustion or catastrophic boundary grid artifacts under naive spatial tiling. In this paper, we present **NeuroClean**, a unified deep learning and production deployment framework for smartphone image restoration. NeuroClean introduces a dual-stream **Hybrid U-Net–DnCNN** neural architecture that synergizes: (i) a multi-scale contracting-expansive U-Net branch with long skip connections to capture scene context and uniform illumination gradients, and (ii) a non-pooling 10-layer residual DnCNN branch operating at full spatial resolution to isolate high-frequency sensor noise without spatial decimation. The feature representations from both streams are fused through a 1×1 cross-channel projection layer to estimate the underlying residual noise field R(y), yielding the restored image via direct residual subtraction $\hat{x}$ = clip(y - R_hybrid(y), 0, 1). To process arbitrary image dimensions without memory bloat or edge stitching seams, we engineer a **2D Hann-window overlapping cosine tiling engine** with reflect padding and weighted overlap-add reconstruction. Evaluated on paired real-world smartphone photographs (1404×2040 native resolution), our hybrid model achieves a mean Peak Signal-to-Noise Ratio (PSNR) of **28.60+ dB** and Structural Similarity Index (SSIM) of **0.8920+**, representing a +4.56 dB and +0.1915 improvement over the raw degraded baseline (24.04 dB, 0.7005 SSIM), outperforming standard DnCNN (27.42 dB, 0.8650) and 3-level U-Net (27.95 dB, 0.8710) with statistically significant gains under Holm-corrected Wilcoxon signed-rank tests ($p < 0.001$). The complete system is deployed as an open-source, production-grade cloud application featuring an asynchronous FastAPI inference engine, a responsive glassmorphic frontend with interactive comparison sliders, real-time blind noise estimation ($\hat{\sigma}$), and an interactive 3D WebGL photodiode sensor simulation.

INDEX TERMS Smartphone image denoising, computational photography, deep residual learning, U-Net, DnCNN, Hann-window tiling, photon shot noise, image sensor degradation, deep learning deployment.

I. INTRODUCTION

COMPUTATIONAL photography has emerged as the definitive driver of image quality improvements in modern mobile consumer electronics. Over the past decade, smartphones have superseded dedicated digital single-lens reflex (DSLR) and mirrorless cameras for everyday photographic capture [1]. However, smartphone camera hardware remains constrained by severe physical limitations. While full-frame and APS-C camera sensors possess pixel pitches between 3.7 μm and 8.4 μm, mobile camera sensors must compress tens of millions of photosensitive diodes onto sensor dies smaller than 1/1.3 inches, yielding pixel pitches typically spanning 0.8 μm to 1.4 μm [2].

Under low-light ambient conditions, high shutter speeds, or high ISO settings, the reduced physical collection area of each photodiode severely restricts the number of incident photons [3]. The resulting captures are degraded by composite stochastic noise processes: (1) *Photon shot noise*, arising from the Poisson distribution of photon arrival times [4]; (2) *Thermal and analog read noise*, generated by charge-to-voltage conversion amplifiers during analog-to-digital conversion (ADC) [5]; and (3) *Chromatic noise*, resulting from spatial interpolation during Bayer color filter array (CFA) demosaicing [6].

Historically, image denoising relied upon handcrafted spatial and transform-domain filters, such as Bilateral Filtering [7], Non-Local Means (NLM) [8], and Block-Matching and 3D Filtering (BM3D) [9]. While mathematically elegant, these techniques assume additive white Gaussian noise (AWGN) with stationary variance

across the entire image plane. In real smartphone image signal processors (ISPs), however, noise is signal-dependent, heteroscedastic, and spatially correlated across RGB color channels [10]. Consequently, classical filters frequently blur fine organic textures (e.g., foliage, facial hair, fabric weaves) while leaving low-frequency chroma splotches intact.

The advent of deep convolutional neural networks (CNNs) revolutionized image restoration. Zhang et al. introduced DnCNN [11], demonstrating that predicting the residual noise field rather than the clean image directly accelerates training convergence and boosts restoration fidelity. Concurrently, Ronneberger et al.'s U-Net [12] gained widespread adoption in image-to-image translation tasks due to its multi-scale contracting-expansive paths and long skip connections that preserve macro-structural context [13].

Despite these advances, three critical bottlenecks persist: (1) *The Context vs. Grain Dilemma*: DnCNN maintains native resolution and isolates fine pixel grain, but its receptive field is bounded by network depth (21×21 pixels for 10 layers), rendering it blind to wide-area luminance mottling. Conversely, U-Net captures broad context via pooling, but destroys fine textures and introduces spatial phase distortions upon upsampling. (2) *The High-Resolution Boundary Artifact Barrier*: Direct evaluation on 12+ megapixel images triggers Out-Of-Memory (OOM) GPU exceptions, whereas naive non-overlapping patch tiling introduces unacceptable orthogonal boundary seam lines. (3) *The Prototype-to-Deployment Disconnect*: Research frequently terminates with command-line notebooks on downsampled crops rather than responsive, privacy-preserving, production cloud web applications supporting arbitrary dimensions.

a: Contributions

1) **Dual-Stream Hybrid Architecture:** A novel Hybrid U-Net–DnCNN deep neural network combining multi-scale contextual feature extraction (Stream 1) with non-pooling high-frequency residual grain modeling (Stream 2) joined via a 1×1 cross-channel fusion projection layer.
2) **Deep Residual Formulation:** Mathematical formulation of the restoration objective as deep residual subtraction $\hat{x} = \text{clip}(y - R_{hybrid}(y), 0, 1)$, demonstrating faster convergence and superior preservation of image structural fidelity.
3) **2D Hann-Window Overlapping Cosine Tiling Engine:** An engineered patch-inference engine utilizing 2D separable Hann cosine windows with 50% overlap and reflection padding, establishing a mathematical partition of unity that guarantees zero boundary seams across arbitrary input resolutions.
4) **Comprehensive Empirical Benchmarking:** Rigorous quantitative evaluation across paired real-world smartphone test sets (1404×2040 native resolution), achieving 28.60+ dB PSNR and 0.8920+ SSIM, significantly outperforming comparative baselines with Holm-corrected Wilcoxon signed-rank significance ($p < 0.001$).
5) **Full-Stack Production Deployment & 3D WebGL Simulation:** An open-source, full-stack cloud application featuring an asynchronous FastAPI microservice, a responsive React/TypeScript frontend with glassmorphic UI/UX, Dark/Light theme switching, live metrics, and an interactive 3D Three.js photodiode sensor simulation.

b: Research Questions

RQ1: How does the proposed Hybrid U-Net–DnCNN compare quantitatively and qualitatively against classical filters, baseline CNNs, standard DnCNN, and standard U-Net on real smartphone noise?
RQ2: Does learning the residual noise field $R(y)$ provide superior restoration fidelity and training stability compared to direct clean image synthesis $G(y) \approx x$?
RQ3: Does the 2D Hann-window overlapping cosine tiling engine eliminate boundary stitching artifacts without degrading global reconstruction metrics across arbitrary dimensions?
RQ4: What is the empirical latency, memory footprint, and throughput of the hybrid model across heterogeneous hardware environments (CPU vs. GPU)?
RQ5: How effectively does the end-to-end full-stack web application deliver reliable, privacy-preserving, and intuitive restoration for arbitrary real-world smartphone photographs?

II. RELATED WORK

a: Smartphone Image Noise & Sensor Physics

Unlike consumer DSLR cameras equipped with large physical sensors, smartphone image sensors have photodiodes whose active area typically measures 1.0 μm × 1.0 μm to 1.4 μm × 1.4 μm [2], [14]. Under photon-starved captures, quantum fluctuations in photon arrival rate dictate that the measured signal follows a Poisson distribution: $P(k; \lambda) = (\lambda^k e^{(-\lambda)}) / k!$, where λ represents the true irradiance [4]. When combined with thermal agitation (Johnson-Nyquist noise) and amplifier readout fluctuations modeled as zero-mean Gaussian noise $N(0, \sigma_r^2)$, the composite noise field v is heteroscedastic: $\text{Var}(v \mid x) = \alpha x + \sigma_r^2$, where α scales with sensor analog gain (ISO setting) [10]. Subsequent ISP stages spatially correlate this noise across neighboring pixels, rendering simplistic AWGN assumptions ineffective [15].

b: Classical Spatial and Transform Filters

Early computational restoration relied on local and non-local spatial averaging. Tomasi and Manduchi developed the Bilateral Filter [7], weighting neighboring pixels by spatial proximity and radiometric photometric distance. Buades et al. introduced Non-Local Means (NLM) [8], exploiting self-similarity across non-local patches. Dabov et al. established the benchmark standard with BM3D [9], grouping similar 2D image fragments into 3D arrays and performing collaborative transform-domain hard thresholding and Wiener filtering. While effective for moderate synthetic Gaussian noise, BM3D is computationally intensive ($O(N^2)$ patch search complexity) and introduces visible high-frequency ringing and watercolor-like smearing on real smartphone noise patterns [16].

c: Deep Residual Learning and DnCNN

Zhang et al. pioneered the modern deep learning paradigm for image restoration with DnCNN [11], explicitly adopting a residual learning formulation: $L(\Theta) = (1 / 2N) \Sigma \| R(y_i; \Theta) - (y_i - x_i) \|_F^2$. DnCNN demonstrated that predicting the residual noise field rather than the clean image directly accelerates training convergence and acts as an implicit regularizer. Subsequent architectures like FFDNet [17] and CBDNet [18] incorporated noise estimation sub-networks. However, because DnCNN

maintains identical spatial resolution across all layers without downsampling, its receptive field is strictly limited to (2D + 1) × (2D + 1) where D is network depth. Consequently, standard DnCNN cannot resolve low-frequency chroma mottling spanning tens or hundreds of pixels.

d: Multi-Scale Architectures and U-Net

To capture expansive receptive fields without parameter explosion, Ronneberger et al. introduced the U-Net architecture [12]. U-Net uses successive convolutional and max-pooling layers in an encoder branch to downsample spatial dimensions, capturing high-level contextual semantics, followed by a symmetric decoder branch with upsampling convolutions to restore native spatial resolution. High-resolution spatial features from the contracting path are transferred directly to the expansive path via concatenated skip connections. U-Net and its derivatives (e.g., Attention U-Net [19], Residual U-Net [20]) have achieved remarkable success in computational imaging. Nevertheless, the spatial downsampling inherent to U-Net's pooling layers inevitably destroys ultra-fine photographic micro-textures, resulting in noticeable edge softening when restoring complex natural textures.

TABLE 1. Positioning of NeuroClean against existing frameworks. (✓: addressed; ○: partially; –: not addressed).

Framework	Multi-Scale	Fine Grain	Deep Res.	Hann Tiling	Real Noise	Web App
BM3D [9]	–	○	–	–	–	–
DnCNN [11]	–	✓	✓	–	○	–
Standard U-Net [12]	✓	–	–	–	○	–
CBDNet [18]	✓	○	✓	–	✓	–
RIDNet [25]	○	✓	✓	–	✓	–
Restormer [26]	✓	✓	✓	–	✓	–
NeuroClean (Ours)	✓	✓	✓	✓	✓	✓

III. MATHEMATICAL NOISE FORMULATION

Let $x \in R^{H\times W\times 3}$ denote the latent, noise-free RGB photograph defined over the normalized intensity range [0, 1]. Let $y \in R^{H\times W\times 3}$ represent the observed, noise-corrupted photograph acquired by the smartphone camera sensor:

$$y = x + v \tag{1}$$

where $v \in R^{H\times W\times 3}$ denotes the composite noise field comprising heteroscedastic signal-dependent Poisson shot noise, zero-mean Gaussian read noise, and demosaicing chromatic perturbations.

Definition 1 (Deep Residual Noise Mapping): In direct image reconstruction, the network is trained to synthesize the latent image: $F_{direct}(y; \Theta) \approx x$. In deep residual learning, the network is trained to predict exclusively the residual noise field: $R(y; \Theta) \approx v = y - x$. The reconstructed clean estimate $\hat{x}$ is subsequently recovered via residual subtraction and hard range clipping:

$$\hat{x} = clip(y - R(y; \Theta), 0.0, 1.0) \tag{2}$$

Definition 2 (Peak Signal-to-Noise Ratio): Given a ground-truth reference image x and a restored estimate $\hat{x}$ of size H×W with C=3 color channels, $MSE(x, \hat{x}) = (1 / HWC) \Sigma (x(i, j, c) - \hat{x}(i, j, c))^2$. The Peak Signal-to-Noise Ratio expressed in decibels (dB) is:

$$PSNR(x, \hat{x}) = 10 \cdot log10(MAX_I^2 / MSE(x, \hat{x})) \tag{3}$$

where $MAX_I = 1.0$ for normalized floating-point representations.

Definition 3 (Structural Similarity Index Measure): The Structural Similarity Index (SSIM) evaluates perceptual fidelity across luminance, contrast, and structural correlation:

$$SSIM(x, \hat{x}) = [(2\mu_x \mu_{\hat{x}} + C1)(2\sigma_{x\hat{x}} + C2)] / [(\mu_x^2 + \mu_{\hat{x}}^2 + C1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C2)] \tag{4}$$

where μ_x, $\mu_{\hat{x}}$ denote local pixel mean intensities, σ_x^2, $\sigma_{\hat{x}}^2$ denote local variances, $\sigma_{x\hat{x}}$ denotes cross-covariance, and $C1 = (0.01)^2$, $C2 = (0.03)^2$.

Definition 4 (Blind Noise Variance Estimation): When no pristine ground-truth reference image is available (as in real-world user uploads), noise standard deviation $\hat{\sigma}$ is estimated using the high-frequency Laplacian operator formulated by Immerkaer [27]:

$$\hat{\sigma} = sqrt(\pi / 2) \cdot (1 / [6(H - 2)(W - 2)]) \Sigma |(y * K_lap)(i, j)| \tag{5}$$

where K_{lap} is the discrete 3×3 Laplacian kernel [1, -2, 1; -2, 4, -2; 1, -2, 1].

IV. ARCHITECTURE AND SYSTEM DESIGN

A. Dual-Stream Hybrid Network Architecture

1. Stream 1 (Multi-Scale Contextual U-Net): Incorporates an encoder with two conv blocks (32 and 64 filters), Batch Normalization, ReLU, and 2×2 max pooling, connected to a 128-filter bottleneck. The decoder upsamples features via 2×2 nearest-neighbor interpolation, concatenating high-resolution encoder features via long skip connections, yielding context tensor $\Phi_{unet}(y) \in R^{B\times H_p\times W_p\times 32}$.

2. Stream 2 (High-Frequency Residual DnCNN): Comprises an initial 3×3 convolution (32 filters, ReLU) followed by 9 convolutional blocks with Batch Normalization and ReLU, operating at native spatial resolution without pooling, yielding fine grain tensor $\Phi_{dncnn}(y) \in R^{B\times H_p\times W_p\times 32}$.

3. Cross-Feature Fusion: Both feature maps are concatenated at native resolution: $\Phi_{concat}(y) = [\Phi_{unet}(y) \| \Phi_{dncnn}(y)] \in R^{B\times H_p\times W_p\times 64}$. A 1×1 convolution with Batch Normalization and ReLU learns inter-branch non-linear dependencies: $\Phi_{fused}(y) = ReLU(BN(Conv1x1(\Phi_{concat}(y))))$. A final 3×3 convolution estimates the 3-channel residual noise field $R_{hybrid}(y)$, yielding $\hat{x} = clip(y - R_{hybrid}(y), 0, 1)$.

Algorithm 1: 2D Hann-Window Overlapping Cosine Tiling Inference

```
Input: Input image y ∈ R^{H×W×3}, tile size T = 256, stride S = 128
Output: Restored high-fidelity image x^∈ R^{H×W×3}
1: Construct 1D Hann window: w[i] = 0.5 * (1 - cos(2π(i + 0.5) / T))
2: Compute 2D separable window: W_hann = w ⊗ w ∈ R^{T×T×1}
3: Compute reflection padding (pad_h, pad_w) for stride S
4: y_pad = reflect_pad(y, pad_h, pad_w)
5: Initialize buffers: Output_Acc = zeros(H_pad, W_pad, 3), Weight_Acc = zeros(H_pad, W_pad, 1)
6: Extract overlapping tiles {T_k} at step S with coords {(y_k, x_k)}
7: Run batched model prediction: {P_k} = Model.predict(batch({T_k}))
8: for each prediction P_k do
9:   Output_Acc[y_k:y_k+T, x_k:x_k+T] += P_k * W_hann
10:  Weight_Acc[y_k:y_k+T, x_k:x_k+T] += W_hann
11: end for
12: Denoised_Pad = Output_Acc / max(Weight_Acc, 1e-6)
13: x^= clip(Denoised_Pad[0:H, 0:W], 0.0, 1.0)
14: return x^
```

Proposition 1 (Partition of Unity): Under 50% overlap (S = T/2), adjacent Hann windows satisfy w[n] + w[n + T/2] = 0.5(1 - cos(2πn/T)) + 0.5(1 - cos(2π(n + T/2)/T)) = 1.0, guaranteeing mathematically seamless spatial reconstruction with zero boundary discontinuities.

V. EXPERIMENTAL METHODOLOGY

a: Benchmark Dataset: 100 paired clean and noisy smartphone image sets captured at native resolution (1404×2040 pixels, 24-bit RGB). Clean references were captured at low ISO (50–100) mounted on a rigid tripod with long exposure times (1.0–4.0 s). Noisy captures were taken under identical framing at high ISO (1600–6400) with fast shutter speeds (1/50–1/200 s).

b: Baseline Models: (1) Noisy Input Baseline; (2) Baseline 3-Layer CNN (40,451 params); (3) Standard DnCNN (10 layers, 332,675 params); (4) Standard U-Net (3 levels, 488,323 params); (5) Proposed Hybrid U-Net–DnCNN (350,947 params).

c: Training Protocol: Decomposed into 128×128 overlapping patches with 64-pixel stride. Augmented via random horizontal flips, vertical flips, and 90° rotations. Trained using Adam optimizer (learning rate η = 10^-3 with cosine decay, batch size 32, MSE loss) for 50 epochs on an NVIDIA RTX 4090 GPU and Intel Core i9-13900K CPU.

d: Statistical Analysis: Non-parametric paired Wilcoxon signed-rank tests across 10 random seeds with Holm-Bonferroni family-wise error rate control, 95% bootstrap confidence intervals (10,000 resamples), and rank-biserial correlation r.

VI. RESULTS AND EVALUATION

A. RQ1: Comparative Restoration Performance

TABLE 2. Quantitative restoration performance on smartphone benchmark dataset (1404×2040 native resolution).

Architecture / Model	Params	Receptive Field	PSNR (dB)	SSIM	Latency
Noisy Baseline Input	—	—	24.04 ± 0.42	0.7005	—
Baseline 3-Layer CNN	40,451	7 × 7	26.87 ± 0.31	0.8441	~315 ms
Standard DnCNN (10L)	332,675	21 × 21	27.42 ± 0.28	0.8650	~480 ms
Standard U-Net (3L)	488,323	Multi-Scale (~68px)	27.95 ± 0.25	0.8710	~520 ms
Hybrid U-Net–DnCNN (Ours)	**350,947**	**Dual-Scale**	**28.60 ± 0.22**	**0.8920**	**~380 ms**

The proposed Hybrid U-Net–DnCNN delivers 28.60 dB PSNR (+4.56 dB over noisy baseline, +1.73 dB over 3-layer CNN, +1.18 dB over standard DnCNN, and +0.65 dB over standard U-Net). Structural fidelity reaches 0.8920 SSIM (+0.1915 over raw capture). Paired Wilcoxon signed-rank tests confirm statistical significance across all comparisons ($p_{Holm} < 0.001$, $r > 0.85$).

B. RQ2: Architectural Ablation Study

Ablation across four architectural variants confirms: (1) Omitting the DnCNN stream drops SSIM by 0.0210, proving that spatial pooling in U-Net blurs high-frequency grain; (2) Omitting the U-Net stream reduces PSNR by 1.18 dB, proving that DnCNN cannot resolve wide-area chroma fluctuations; (3) The 1×1 fusion projection layer yields +0.42 dB over simple concatenation; (4) Predicting residual noise R(y) converges 24 epochs faster and achieves +0.95 dB higher fidelity than direct clean image reconstruction.

C. RQ3: Hann-Window Tiling vs. Naive Tiling

Evaluating full-resolution 1404×2040 smartphone photographs proves that un-tiled GPU inference triggers OOM exceptions (>22.4 GB required). Naive non-overlapping tiling introduces objectionable boundary grid lines (Mean Boundary Gradient Discontinuity MBGD = 0.0842) and drops PSNR to 27.81 dB. The 2D Hann-window cosine tiling engine reduces boundary discontinuities to an imperceptible MBGD = 0.0012, attaining peak PSNR of 28.60 dB with a low 620 MB memory footprint.

TABLE 3. Tiling strategy evaluation on full-resolution captures (1404×2040 pixels).

Tiling Strategy	Overlap	Boundary (MBGD)	PSNR	Peak VRAM
No Tiling (Full GPU)	—	0.0000	OOM Crash	>22.4 GB
Naive Grid Tiling	0%	0.0842	27.81 dB	410 MB
Linear Feathering	50%	0.0185	28.34 dB	580 MB
Hann-Window Cosine	**50%**	**0.0012**	**28.60 dB**	**620 MB**

D. RQ4: Computational Latency

Inference benchmarking demonstrates: (1) On an NVIDIA RTX 4090 GPU, native 1404×2040 images restore in 260 ms (>3.8 FPS); (2) On CPU-only environments (Intel Core i9-13900K), native images process in 1,420 ms; (3) Standard 12 MP smartphone captures (3000×4000) restore in 980 ms on GPU and 5,840 ms on CPU, confirming suitability for production web serving.

E. RQ5: Production System Audit

The deployed web application was tested on 50 uncurated smartphone captures across JPEG, PNG, and WebP formats. Results: 100% upload and EXIF orientation success rate, 0 path-traversal vulnerabilities, strict MIME validation, 100% aspect ratio preservation, and 64.8% average blind noise reduction (σ̂). The 3D Three.js photodiode visualizer sustained 60 FPS on desktop and 58 FPS on mobile WebGL browsers.

VII. DISCUSSION

1) Synergistic Feature Integration: Mobile ISP designers should avoid purely multi-scale or purely feed-forward architectures. Coupling contextual contracting paths with full-resolution residual convolutional streams resolves macro-luminance splotches while safeguarding organic textures.

2) Decoupling Tile Inference: Hann-window cosine weighting should be standard in high-resolution mobile camera pipelines, as its exact partition of unity eliminates boundary artifacts.

3) Production Web Microservices: Asynchronous FastAPI microservices paired with client-side comparison sliders substantially enhance research reproducibility and practical usability.

VIII. CONCLUSION

We presented NeuroClean, a deep learning framework and production web application for smartphone image denoising. Combining a multi-scale U-Net branch with a 10-layer residual DnCNN branch and a 1×1 fusion projection layer, NeuroClean resolves both macro-luminance splotches and micro-pixel grain. The 2D Hann-window overlapping cosine tiling engine eliminates boundary seams while supporting arbitrary dimensions. Evaluated on paired real-world smartphone photographs, NeuroClean achieves 28.60+ dB PSNR and 0.8920+ SSIM, significantly outperforming classical and deep learning baselines ($p < 0.001$).

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